

Milestone 4 - System Prototype Notebook Copy

Project: NLP-assisted job opportunity matching for MSBA international students

Team GitHub notebook URL:

https://github.com/Kongbai815/job-matching-nlp/blob/main/MSBA_Job_Matching_Milestone4_System_Prototype.ipynb

1. End-to-End Architecture

Input advisor query -> normalize text -> retrieve similar real postings from the 80,000-row training corpus -> apply a transparent role-fit rubric -> generate a grounded recommendation that cites source fields and flags missing authorization evidence.

2. Real Input and Output

Input query: Find entry-level US analytics or business analyst opportunities for MSBA international students with SQL, Python, Excel, Tableau, or Power BI skills. Flag cases where authorization evidence is missing.

Output headline: Review 3 of the retrieved postings first; they match analytics role-fit signals.

Rank	Posting	Company	Label
1	Entry level / Busines Data Analyst (Re	Lumos Stratgy	high_fit
2	Entry Level Data Analyst	SkiHomeRealty	high_fit
3	Entry Level Business Analyst/Data Anal	Alipro	high_fit
4	Entry Level Data Analyst - US Army (13	United States Army	unclear
5	Business Analyst in BI Analytics Guild	Swedbank	unclear

3. Evaluation

System	Rows	Accuracy	Macro F1
M2 full validation baseline	20,000	79.3%	78.5%
Same-subset hashed centroid baseline	4,000	79.7%	79.1%
M4 retrieval + grounded prototype	4,000	80.0%	80.1%

Grounding check: retrieval success rate 100.0%; support hit@5 90.0%. Authorization language is not generated unless source evidence exists.

4. Risk

The main risk is hallucinating CPT, OPT, or sponsorship eligibility. The prototype mitigates this by using template generation only and by explicitly marking authorization evidence as unavailable in the public data. The advisor remains the final reviewer.